

Hands-on Landscape-Based Ecological Risk Assessment Using the Open-Source xLandscape Modelling Framework

SETAC Europe, 17 May 2026

"A SETAC Europe **training course on landscape modelling** has so much potential to be **engaging, playful, and genuinely memorable**, especially because **the topic sits at the intersection of ecology, spatial thinking, and systems dynamics.**" (Copilot March'26)



Wetterau, Hessian, Germany

Introductory Remarks

SETAC Europe, 17 May 2026

This is the first training course on landscape modelling using the xLandscape framework.

To make up for any gaps in our early-stage preparation, please reach out with questions or roadblocks as you begin using xLandscape in your projects.

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Wetterau, Hessian, Germany

Welcome

I'm Thorsten Schad

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Biologist, with a background in physics, math and informatics
27+ years in environmental and landscape modelling,
esp. in risk assessment of pesticides

Co-creator of xLandscape

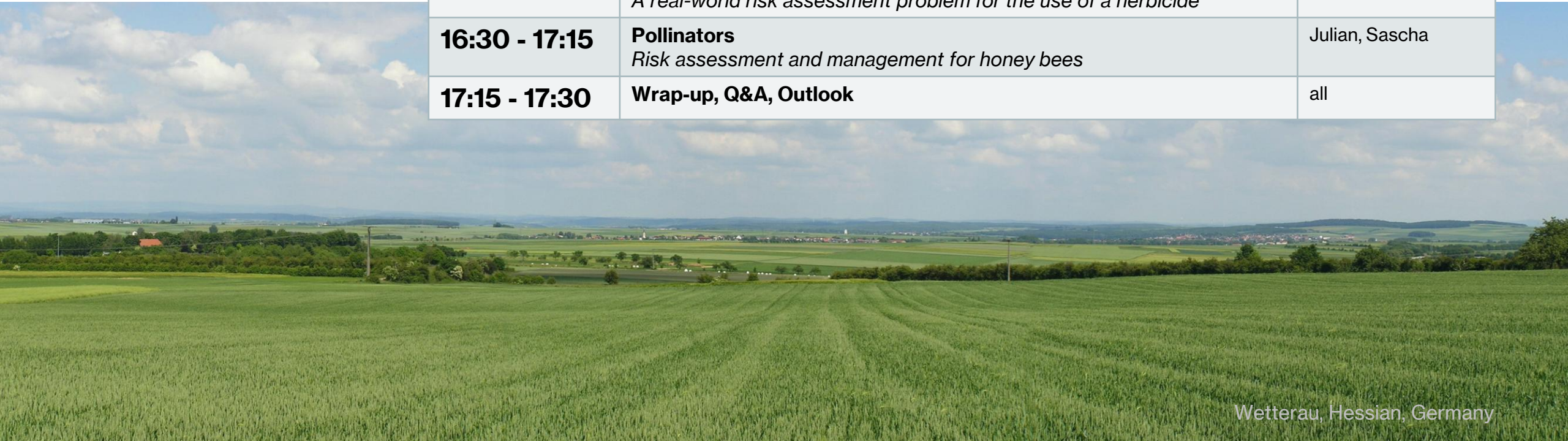
Passionate about cultivated landscapes, balancing human
welfare, natural ecosystems and biodiversity



Content

13.30 – 17.30

13:30 - 13:50	Introduction and Fundamentals <i>Landscape Modelling in risk assessment of pesticides</i>	Thorsten
13:50 - 14:15	Introduction to xLandscape <i>Motivation, concepts, design</i>	Thorsten, Sascha
14:15 - 15:15	Aquatic Risk of Invertebrates <i>A real-world risk assessment problem for the use of an insecticide</i>	Thorsten
15:15 - 15:30	Coffee Break	
15:30 - 16:30	Aquatic Risk of Water Plants <i>A real-world risk assessment problem for the use of a herbicide</i>	Sebastian
16:30 - 17:15	Pollinators <i>Risk assessment and management for honey bees</i>	Julian, Sascha
17:15 - 17:30	Wrap-up, Q&A, Outlook	all



The Rise of AI

A game changer for

xLandscape development

Analysis, visualization, reporting

User Interface

Scenario development

Documentation, Training

→ Consequences on training take home

Technical implementation, model use (User Interface)

Range of application

Model Outcome, Analysis, Communication

Concepts, esp. modularity. Architectures.

Motivation for Landscape Modelling. Strategies.

Regulatory framework.

low AI impact
↑
high

The screenshot displays a development environment with three main components:

- Git Commit History:** A table showing recent commits with columns for Graph, Description, Date, Author, and Commit. The most recent commit is titled "feat(controlpanel): make analysis exposure model run-aware" and was committed by bde38b05 on 13 Apr 2026.
- Chat Window (GitHub Copilot):** A chat interface showing a conversation about enhancing analysis functionality in the control panel. It includes a commit message and a list of included files.
- Terminal Window:** A terminal window showing the output of a command, likely related to the development process.

